Unit: 6 Unit 7: Factoring
Topic: Finding Common Factors

Name: ________________
Date: ___________

Foundation (Jalapeno)

Q1. What is the greatest common factor of 12 and 18?
(A) 1
(B) 2
(C) 3
(D) 6
(E) 9

Q2. A basketball player scored 15 points in the first quarter and 20 points in the second quarter. What is the greatest common factor of these two numbers?
(A) 1
(B) 2
(C) 5
(D) 10
(E) 15

Q3. Factor out the greatest common factor from the expression $8x + 12y$
(A) $2(4x + 6y)$
(B) $4(2x + 3y)$
(C) $8(x + y)$
(D) $12(x + y)$
(E) $2x + 3y$

Q4. A cyclist rode 24 miles in the morning and 30 miles in the afternoon. What is the greatest common factor of these two distances?
(A) 1
(B) 2
(C) 3
(D) 6
(E) 12

Q5. Simplify the expression $9x^2+12x$ by factoring out the greatest common factor
(A) $3x(3x + 4)$
(B) $3x(3x + 2)$
(C) $3(3x^2 + 4x)$
(D) $3x(3x + 1)$
(E) $3(3x^2 + 2x)$

Q6. A swimmer completed 18 laps in the first heat and 24 laps in the second heat. What is the greatest common factor of these two numbers?
(A) 1
(B) 2
(C) 3
(D) 6
(E) 12

Q7. Factor out the greatest common factor from the expression $16x^2 + 20x$
(A) $4(4x^2 + 5x)$
(B) $4x(4x + 5)$
(C) $4(4x^2 + 4x)$
(D) $2x(8x + 10)$
(E) $2x(8x + 5)$

Q8. A golfer scored 20 points on the first hole and 25 points on the second hole. What is the greatest common factor of these two numbers?
(A) 1
(B) 2
(C) 5
(D) 10
(E) 15

Open Ended Questions (FRQ)

Q9. Simplify the expression $12x^3 + 18x^2$ by factoring out the greatest common factor. Show all your work.

Q10. A football player ran 24 yards in the first quarter and 30 yards in the second quarter. What is the greatest common factor of these two distances? Use this information to simplify the expression $8x + 10y$, where x represents the distance run in the first quarter and y represents the distance run in the second quarter.

Chef's Tips

- Use factoring to simplify expressions
- Identify the greatest common factor of two or more numbers
- Apply factoring to real-world problems involving sports and fitness